breaking[24] and is one of the most important ideas to emerge in the later decades of twentieth-century theoretical physics. Let's see why.

The Higgs Ocean and the Origin of Mass

If a Higgs field has a nonzero value—if we are all immersed in an ocean of Higgs field—then shouldn't we feel it or see it or otherwise be aware of it in some way? Absolutely. And modern theory claims we do. Take your arm and swing it back and forth. You can feel your muscles at work driving the mass of your arm left and right and back again. If you take hold of a bowling ball, your muscles will have to work harder, since the greater the mass to be moved the greater the force they must exert. In this sense, the mass of an object represents the resistance it has to being moved; more precisely, the mass represents the resistance an object has to changes in its motion—to accelerations—such as first going left and then right and then left again. But where does this resistance to being accelerated come from? Or, in physics-speak, what gives an object its inertia?

In Chapters 2 and 3 we encountered various proposals Newton, Mach, and Einstein advanced as *partial* answers to this question. These scientists sought to specify a standard of rest with respect to which accelerations, such as those arising in the spinning-bucket experiment, could be defined. For Newton, the standard was absolute space; for Mach, it was the distant stars; and for Einstein, it was initially absolute spacetime (in special relativity) and then the gravitational field (in general relativity). But once delineating a standard of rest, and, in particular, specifying a benchmark for defining accelerations, none of these scientists took the next step to explain *why* objects resist accelerations. That is, none of them specified a mechanism whereby an object acquires its mass—its inertia— the attribute that fights accelerations. With the Higgs field, physicists have now suggested an answer.

The atoms that make up your arm, and the bowling ball you may have picked up, are all made of protons, neutrons, and electrons. The protons and neutrons, experimenters revealed in the late 1960s, are each composed of three smaller particles known as quarks. So, when you